

# Doctor–Patient Interaction Platform Backend Development Documentation

## 1. Project Overview

This document explains the complete backend architecture, workflows, APIs, database design, and system logic for the Doctor–Patient & Care Provider Interaction Platform. It is intended for backend developers to clearly understand what needs to be built and how the system should function.

## 2. User Roles

**Patient:** Can register, login, book appointments, view medical records.

**Provider:** Doctor, Nurse, or Care Taker. Can manage profile, accept/reject appointments, and add medical notes.

**Admin:** Full control over users, providers, appointments, payments, and system configuration.

## 3. Recommended Tech Stack

**Backend:** Node.js (Express.js)

**Database:** MongoDB

**Authentication:** JWT + OTP (SMS/Email)

**File Storage:** AWS S3 / Cloudinary

**Notifications:** SMS (Twilio), Email (Nodemailer)

**Hosting:** AWS / DigitalOcean

## 4. Authentication & Authorization Flow

- User registers with mobile/email.
- OTP is generated and verified.
- JWT token is issued after verification.
- Role-Based Access Control (RBAC) restricts APIs per role.

## 5. Database Schema (High Level)

**User Collection:** name, email, mobile, role, isVerified, createdAt

**Provider Collection:** userId, category, specialization, fees, availability, status

**Appointment Collection:** patientId, providerId, date, time, status, notes

**MedicalRecord Collection:** patientId, providerId, remarks, documents

**Payment Collection:** appointmentId, amount, status, payoutStatus

## 6. Core API Modules

**Auth APIs:** Register, Login, Verify OTP, Logout

**Patient APIs:** Profile, Book Appointment, View History

**Provider APIs:** Profile Update, Appointment Actions, Add Notes

**Admin APIs:** Approvals, User Management, Reports

## 7. Appointment Workflow

1. Patient searches provider.
2. Appointment request is created (Pending).
3. Provider accepts/rejects.
4. Status updates are notified to patient.
5. Consultation notes are added post-visit.

## 8. Admin Workflow

- Admin reviews provider registration.
- Admin approves/rejects provider.
- Admin manages appointments, schedules, and payouts.

## 9. Security & Compliance

- JWT token validation on all protected routes.
- Encrypted sensitive data.
- Strict role-based access.
- Medical data privacy compliance.

## 10. Deployment & Environment Variables

ENV variables include DB\_URI, JWT\_SECRET, OTP\_SERVICE\_KEYS, EMAIL\_CONFIG, STORAGE\_KEYS.

## 11. Future Enhancements

- Video consultation integration.
- AI-based appointment recommendations.
- Multi-language support.